Chapter 9

Magnetic Resonance

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1. Introduction

Unpaired electrons in atomic or molecular species and atoms possessing unpaired nucleons (odd Z-odd A, even Z-odd A, and odd Z-even A) have magnetic dipole moments. These mngnctic moments are associated with spin angular momentum. Magnetic resonance involves the collective macroscopic magnetic properties oi these systems and implies correspondence of an external frequency with a natural frequency of the system.

When a system possessing unpaired electrons or nucleons is placed in a static magnetic field, a number of energy levels is differentiated by Zeeman splittings of the quantum states of the magnetic moment. Transitions of the magnetic moment between those levels are induced with an applied radiation at the resonant frequency. Such transitions are indicated by an energy absorption or emission. Detailed information regarding the environment of the atoms and electrons in the material arc obtained from the intensities and the frequencies of resonance absorption and emission.

Magnetic resoanace of assemblies of unpaired electrons is called either electron paramagnetic resonance (EPR) or electron spin resonance (ESR) . The first experimental evidence for maerosropic ma gnetic resonance transitions between Zeeman levels of bulk $CuCl_2{\cdot}2H_2O$ was given by Zavoisky [1]. Cummcrow and Halliday [2] employed 2.9 gigahertz (GHz) radiation to examine the magnetic resonance in bulk $MnSO_4{\cdot}4H_2O$. The first hyperfine structure EPR measurements were reported by Penrose [3] who doped magnesium ammonium nitrate hexahydrate wi th copper (II).

EPR spectroscopy has been used effectively for investigating transition (d electron) ions, inner transition (*f* electron) ions, stable or unstable mono- and bi-radicals, metal-organic compounds, ground and excitcd state triple molecules, donors and acceptors in semiconductors, radiation-induced defects, metals and paramagnetic impurities in solids. The general theory and the applications of EPR spectroscopy have been described in detail in review articles and textbooks [4-27].

The first observation of bulk nuclear paramagnetism was made by Lasarev and Schubnikov [28] when measured the static magnetization of solid hydrogen. The first successful nuclear magnetic resonance experiment on bulk material were performed indepcndcntly by Purcell, Torrey, and Pound [29] and bv Boch, Hansen, and Packard [30]. Purcell and his coworkers observed resonance absorption **of** protons in a hydrocarbon (paraffin) whereas B!och and hi colleagues found the proton resonance signal in water.

Nuclear magnetic-resonance spectroscopy (NMR) has been applied to many areas in physics and chemistry which include the elucidation of the intermolecular structure in solids, liquids, and gases, the determination of magnetic fields in single crystals, molccular absorption, diffusion coefficients, and electric field effects. A number of review articles and books treat the general theory, experimental techniques and applications [31-51].

Useful literature surveys of the applications, and advances of ESR and NMR have been given [52]. In addition, symposia to cover special topics and advances in NMR and EPR have been published [53-62]. A continuing series on magnetic resonance has been started [63].

2. Magnetic Properties of Nuclei and Electrons

If it is assumed that a charge e, and mass, m, are distributed uniformly on a spherical shell which is spinning about a given axis, a magnetic field symmetrical about this axis will be generated by the rotating charge. The charge e, rotating at $v/2\pi r$ revolutions sec^{-1} generates a current $ev/2\pi r$. Since the rotating charge behavcs as **a** magnetic dipole whose moment, μ, is given by the product of the current i and the loop cross-sectional area, πr^2,

$$\mu = i\pi r^2 = \frac{evr}{2c} \text{ (cmu)}$$

where c represents the velocity of light. The spinning mass generates an angular momentum, p, of magnitude mvr directed along the axis of rotation. Thus the magnetic moment is colinear and directly proportional to the angular momentum [64]

$$\vec{\mu} \alpha \frac{e}{2mc} \vec{p} \qquad (9.1)$$

The electron mass is used when unpaired electrons are involved, whereas when unpaired nucleons are present nuclidic masses are required.

The direction of the vectors is determined by the sign of the charge; however, this model does not explain nuclear moments (e.g., neutron). The nuclear model deficiencies suggest a complex structure involving orbital angular momentum and spin angular momentum coupling analgous to that well known for the electron [65].

The proportionality constant between $\vec{\mu}$ and $\vec{p}$ is designated g for the electron and g_N for nuclei. For free atoms the spectroscopic splitting factor, g, is identical with the Lande g value; however, this identity does not hold for all magnetic molecules as will be seen later. The nuclear g_N value or nuclear spectroscopic splitting factor has a characteristic value for each nuclide.

The angular momentum, p, of any particle or particle system can be expressed in units of $\hbar$ (Planck's constant divided by 2π). The spin is defined as the largest observable value of the time average of the angular momentum component in a specified direction ($\langle p \rangle_{max}/\hbar$ = spin). The electron spin is designated by S whereas the nuclear spin is termed I.

For the nucleus, the general expression for all permitted values of p is $p = m\hbar$, where m, the nuclear magnetic quantum number, has $(2I + 1)$ values I, $I - 1, \ldots 0, \ldots, -I + 1, -I$. The lowest lying energy level is characterized by an I which has a positive integral or half integral value. Then the nuclear magnetic moment is given by

$$\mu = g_N \frac{e\hbar}{2Mc} I = g_N \mu_N I \qquad (9.2)$$

where μ_N, the nuclear magneton, has the value 5.05050 X 10^{-24} erg $gauss^{-1}$ [66] for the proton. The term $g_N(e/2Mc)$ is often replnccd by γ, the magnetogyric ratio; hcncc, $\mu = \gamma\hbar I$. The usual nuclear property tables [31, 32, 34-36, 40, 42, 50] list values of μ in nuclear magnetons and I in $\hbar$ units. The magnetic momcnt of the proton is 2.79277 $\pm$ 0.00007 nuclear magnetons [66, 67J.

Nuclides having spins ≥ 1 do not always have a spherical nuclear charge. The charge asphericity is given by the nuclear electric quadrupole moment, Q, which is defined by the relation

$$eQ = \int \rho r^2 (3\cos^2\theta - 1)\, dv \qquad (9.3)$$

where ρ is the charge density, dv is a volume element inside the nucleus located a distance r from the nucleus center and making an angle θ with the nuclear spin axis, c is the protonic charge, and the integration is performed for the state I. Positive eQ values are observed when the charge distribution is elongated along the spin axis (prolate spheroid) whereas a flattened charge distribution (oblate spheroid) along the spin axis is indicated by a ncgntive eQ value. The Q values expressed in barns (10^{-24} cm^2) vary from 10.2 barns for Fr^{187} to -1.7 barns for Ac^{227} [68-73].

The electron magnetic moment is related to the spin quantum number S by $\mu_e = -\gamma_e S\hbar$. Instead of using γ_e the electron magnetic moment is written in terms of the Bohr magneton ($\mu_B = e\hbar/2mc$) and the spectroscopic splitting factor, g_0,

$$\mu_e = -g_0 \mu_B S \qquad (3.4)$$

where μ_B = 0.93732 X 10^{-20} erg $gauss^{-1}$ and g_0 = 2 00232 [66]. The g value is a measure of the contribution of the spin and orbital angular momentum of the electron to the total angular momentum and therefore depends on the electron environment. The spin S refers to the total spin of all unpaired electrons and has a multiplicity 2S + 1 with values ranging from $-S$ to S. The Bohr magncton-proton moment ratio is 657.4181 [66]; hence the Bohr magneton-nuclear moment ratio is 1838.2403.

Static Magnetism. The magnetic induction B and the magnetic field H arc related through classical electromagnetics by

$$B = H + 4\pi M \qquad (9.5)$$

where M, the magnctiration, is the macroscopic magnetic moment per unit volume. The static magnetic susceptibility, χ_0, is defined as the ratio of the magnetization to the magnetic field, H_0,

$$\chi_0 = \frac{M}{H_0} \qquad (9.6)$$

When the susceptibility is positive, paramagnetism results whereas diamagnetism is indicated by a negative susceptibility.

When an unpaired electron is placed in a static magnetic field, tho potential interaction energy, E_i, is given by

$$E_i = -\vec{\mu} \cdot \vec{H} \qquad (9.7)$$

The total angular momentum, J, of the electron is related to its magnetic moment by

$$\mu = g\mu_B J \qquad (9.8)$$

When the applied field direction lies along the z axis, then the resultant Zeeman energy levels are given by

$$E_i = -g\mu_B H_0 m_J \qquad (9.9)$$

whcrc m_J, the magnetic quantum number, has $2J + 1$ multiplicity ranging in value from $-J$ to J. Figure 9.1 shows the energy levels for a $J = \frac{1}{2}$ system.

The probability P_i that an electron within the assembly has a potential energy Et at a temperature T is given by thn Boltzmann relation

$$P_i = \text{constant} \left(\exp - \frac{E_i}{kT}\right) = \frac{\exp(-g\mu_B H_0 m_J/kT)}{\sum_{m_J=-J}^{J} \exp(-g\mu_B H_0 m_J/kT)} \qquad (9.10)$$

Boltzmann statistics are applicable since the magnetization is the property of similar, spatially distinguishable particles [74]. The energy when H_0 is

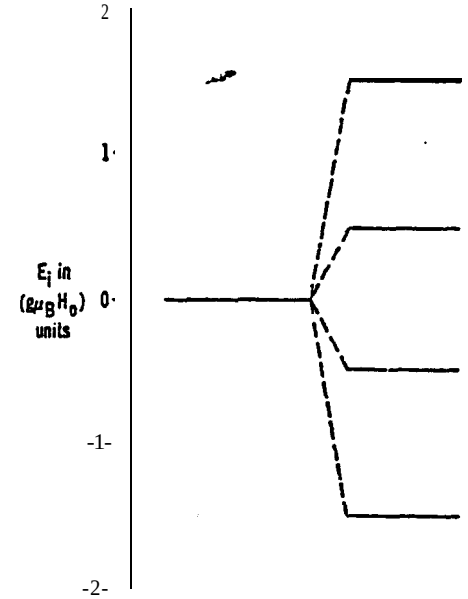


FIG. 9.1. **Zeeman pattern for** $J = \frac{3}{2}$.

parallel to the z axis is

$$E = \frac{N \sum_{m_J=-J}^{J} g\mu_B H_0 m_J \exp(-g\mu_B H_0 m_J/kT)}{\sum_{m_J=-J}^{J} \exp(-g\mu_B H_0 m_J/kT)} \tag{9.11}$$

where N is the number of electrons per unit volume. For the high-temperature approximation $g\mu_B H_0/kT \ll 1$

$$M_z = \frac{\partial E}{\partial H_0} = NJ(J+1)\frac{g^2\mu_B^2 H_0}{3kT} \tag{9.12}$$

Hence the susceptibility, χ_0, is given by the Langevin-Brillouin relation

$$\chi_0 = \frac{NJ(J+1)g^2\mu_B^2}{3kT} \tag{9.13}$$

For conditions where $\frac{g\mu_B H_0}{kT} \gg 1$

then $$M_z = Ng\mu_B J B_J\left(\frac{g\mu_B H_0}{kT}\right) \tag{9.14}$$

where $$B_J\left(\frac{g\mu_B H_0}{kT}\right)$$

is the Brillouin function. [75, 76].

The volume susceptibilitt for unpaired electrons is of the order of 10^{-5}–10^{-6} egs unit. For nonzero spin nuclei the static nuclear susceptibility is given by

$$\chi_0 = \frac{Ng_N^2\mu_N^2 I(I+1)}{3kT} \tag{9.15}$$

For protons in water at room temperature $\chi_0 \approx 3 \times 10^{-10}$ egs unit., which is considerably smaller than the diamagnetic susceptibility (-0.719×10^{-6} egs unit) of water.

3. Classical Magnetic Resonance Description

When the spinning charged particle is placed in an external magnetic field, H_0, the magnetic moment experiences a torque L, tending to align it in the direction of the field. The effect of the torque changes the angular momentum, p,

$$\frac{d\vec{p}}{dt} = \vec{\mu} \times \vec{H} \tag{9.16}$$

or $$\frac{d\vec{\mu}}{dt} = \gamma(\vec{\mu} \times \vec{H})$$

Since the magnetization M is defined by $N\langle\mu\rangle$ where $\langle\mu\rangle$ is the average moment,

$$\frac{d\vec{M}}{dt} = \gamma(\vec{M} \times \vec{H}) \tag{9.17}$$

This equation (9.17) when written in terms **of** the vector components takes the form:

$$\begin{aligned} \frac{dM_x}{dt} &= \gamma(M_y H_z - M_z H_y) \\ \frac{dM_y}{dt} &= \gamma(M_z H_x - M_x H_z) \\ \frac{dM_z}{dt} &= \gamma(M_x H_y - M_y H_x) \end{aligned} \tag{9.18}$$

When the special case of a constant magnetic field in the z direction is considered, the following relations are obtained:

$$\begin{aligned} \frac{dM_x}{dt} &= \gamma M_y H_z \\ \frac{dM_y}{dt} &= -\gamma M_x H_z \\ \frac{dM_z}{dt} &= 0 \end{aligned} \tag{9.19}$$

The solutions of these equations are

$$\begin{aligned} M_x &= M_0 \sin\alpha \cos(\gamma H_0 t + \phi) \\ M_y &= -M_0 \sin\alpha \sin(\gamma H_0 t + \phi) \\ M_z &= M_0 \cos\alpha \end{aligned} \tag{9.20}$$

and represent the magnetization precession about H_z with an angular frcqucncy $\omega = \gamma H_0$ as shown in Fig. 9.2. A special solution $M_z = \pm M_0$ independent of time exists when $\alpha = 0$ or π. The system has a characteristic frequency $\omega_0/2\pi$, termed the Larmor frequency, which equals 42.577 megahertz (MHz) for protons and 27.99 gigahertz (GHz) for electrons when each is placed in a 10-kilogauss (kG) field.

This system may be excited at the Larmor frequency, $\omega_1/2\pi$, with a small oscillating field, H_1, applied perpendicular to H_0. To first order, only the proper circular polarization component will synchronize with the precession frequency [77]; hence a

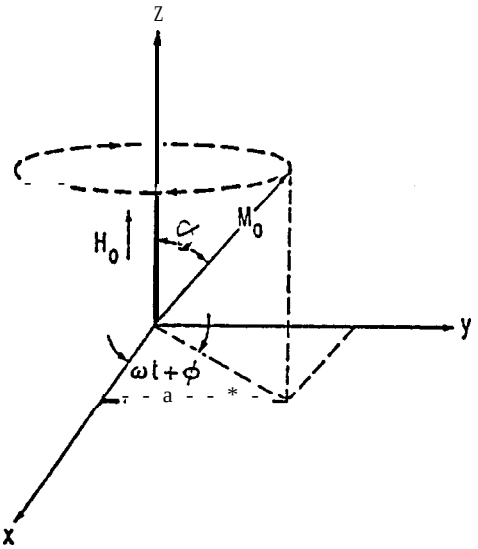


FIG. 9.2. Magnetization precession in a magnetic field.

linear oscillating field, composed of two counterrotating fields, is suitable. To describe this weak field addition, a coordinnte system rotating in the xy plane at a frequency $\omega = \omega_0$ to hold H_1 constant is convenient [78]. Let $\partial M/\partial t$ represent the time variation of M in the rotating coordinate system. Then

$$\frac{d\vec{M}}{dt} = \frac{\partial \vec{M}}{\partial t} - (\vec{M} \times \vec{\omega}) \qquad (9.21)$$

in the laboratory coordinates. If $\vec{\omega}$ is along the z axis (Fig. 9.3), then the components of M are

$$\begin{aligned} \frac{d\vec{M}_x}{dt} &= \frac{\partial \vec{M}_x}{\partial t} - \omega \vec{M}_y \\ \frac{d\vec{M}_y}{dt} &= \frac{\partial \vec{M}_y}{\partial t} + \omega \vec{M}_x \\ \frac{d\vec{M}_z}{dt} &= \frac{\partial \vec{M}_z}{\partial t} \end{aligned} \qquad (9.22)$$

Thus the magnetization M' in the rotating system has the behavior

$$\frac{\partial \vec{M}'}{\partial t} = \gamma \left[\vec{M}' \times \left(\vec{H} + \frac{\vec{\omega}}{\gamma} \right) \right] \qquad (9.23)$$

and the rotating system behaves as a stationary system with an effective magnetic field $[\vec{H} + (\vec{\omega}/\gamma)]$. No effective field exists when $\vec{\omega} = -\vec{\omega}_0 = -\gamma \vec{H}_0$. When H_1 is directed along the x' axis the effective-magnetic-field components arc $H_{x'} = H_1$ and

$$H_z = H_0 - \left(\frac{\omega}{\gamma} \right)$$

The effective magnetic field is directed at an angle

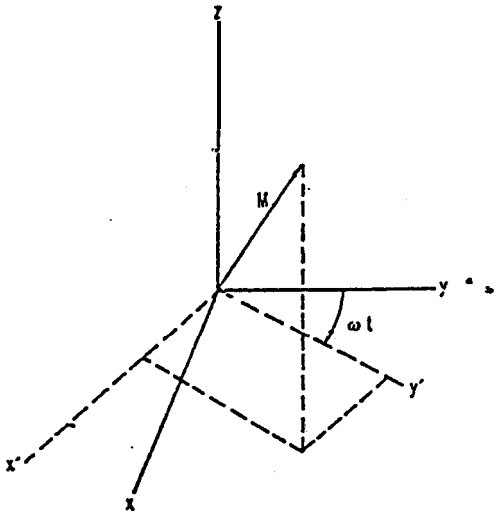


FIG. 9.3. Relation of rotating and laboratory coordinate systems.

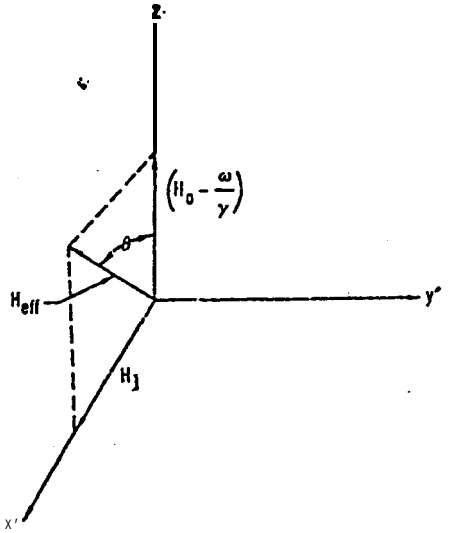


FIG. 9.4. Effective magnetic field in rotating frame.

θ to H_0 when the z axes arc superimposed as shown in Fig. 9.4.

The explicit transformations to the rotating coordinate system for a positive γ are

$$\begin{aligned} H_x &= H_1 \cos \omega t \qquad H_y = -H_1 \sin \omega t \\ M_x &= M_x' \cos \omega t + M_y' \sin \omega t \\ \text{and} \quad M_y &= -M_x' \sin \omega t + M_y' \cos \omega t \end{aligned} \qquad (9.24)$$

Then the time-indcpcnticnt equations for the rotating frame are

$$\begin{aligned} \frac{\partial M_x'}{\partial t} &= \gamma M_y' \left(H_z - \frac{\omega}{\gamma} \right) \\ \frac{\partial M_y'}{\partial t} &= -\gamma M_x' \left(H_z - \frac{\omega}{\gamma} \right) + \gamma H_1 M_z \\ \frac{\partial M_z'}{\partial t} &= -\gamma H_1 M_y' \end{aligned} \qquad (9.25)$$

Now assume at $t = 0$ that M' is parallel to H_0 and that an oscillating field of amplitude H_1 is imposed on the system. At a later time t,

$$\begin{aligned} M_z' &= 2M_0 \sin\theta\cos\theta \sin^2\left(\gamma H_{eff}\frac{t}{2}\right) \\ &= \frac{2M_0\gamma H_1(\omega_0-\omega)}{(\omega_0-\omega)^2+\gamma^2H_1^2}\sin^2\left(\gamma H_{eff}\frac{t}{2}\right) \\ M_y' &= M_0 \sin\theta \sin(\gamma H_{eff} t) \\ &= \frac{M_0\gamma H_1}{[(\omega_0-\omega)^2+\gamma^2H_1^2]^{1/2}}\sin(\gamma H_{eff} t) \\ M_z' &= M_0[\cos^2\theta + \sin^2\theta\cos(\gamma H_{eff} t)] \\ H_1^2 &= M_0\left[1-\frac{2\gamma^2H_1^2}{(\omega_0-\omega)^2+\gamma^2H_1^2}\sin^2\left(\gamma H_{eff}\frac{t}{2}\right)\right] \end{aligned} \tag{9.26}$$

where $\gamma H_{eff} = [(\omega_0 - \omega)^2 + \gamma^2 H_1^2]^{1/2}$ and $\omega_0 = \gamma H_0$. A relation between these quantitirs and the probability that a spin has made a transition between two Zeeman levels is required.

Let $P_{\pm 1/2}$ be the probability of a spin $\frac{1}{2}$ system being in a state with the magnetic quantum number equal to $\pm\frac{1}{2}$. Since $P_{1/2} - P_{-1/2} = M_z'/M_0$ and $P_{1/2} + P_{-1/2} = 1$,

$$\begin{aligned} P_{-1/2} &= \frac{1-(M_z'/M_0)}{2} \\ H_1^2 &= \frac{\gamma^2H_1^2}{(\omega_0-\omega)^2+\gamma^2H_1^2}\sin^2\left(\gamma H_{eff}\frac{t}{2}\right) \end{aligned} \tag{9.27}$$

Now consider the special case $\omega = \gamma H_0 = \omega_0$ then $P_{-1/2} = \sin^2[\gamma H_1(t/2)]$ since $H_{eff} = H_1$ (see Fig. 9.4). This result describes magnetic resonance and indicates that the magnetic moment oscillates between the z and $-z$ directions.

At equilibrium, the relative population of each Zeeman level is given by a Boltzmann distribution $N_j/N_i = \exp -\Delta E_{ij}/kT$ where N_j is the population of the upper level and N_i is the population of the lower level. For the nuclear case where $\Delta m = +1$,

$$\frac{N_j}{N_i} = \exp -\frac{g_N\mu_N H_0}{kT} = 1 - \frac{g_N\mu_N H_0}{kT} \tag{9.28}$$

For $I = \frac{1}{2}$ system at room temperature in

$$H_0 = 10\ kG$$

$g_N\mu_N H_0/kT = 7 \times 10^{-6}$. In the analogous electron case $N_j/N_i = \exp(-g\mu_B H_0/kT) = 1 - g\mu_B H_0/kT$ and $g\mu_B H_0/kT = 3 \times 10\text{-J}$ for a $S = \frac{1}{2}$ system at room tcmpcraturc in H_0 of 10 kG. Thus a net cncrgy absorption will occur during the normal magnetic resonance experiment.

4.* Quantum Mechanical Behavior [79-81]

A quantum mechanical description of magnetic resonance is required to explain coupled spin systems which are not handled by the classical theory. This analysis will be restricted to a nuclear spin of $\frac{1}{2}$ and begins with the Schrodinger equation

$$i\hbar\frac{\partial\psi}{\partial t} = \mathcal{H}\psi \tag{9.23}$$

where ψ is the wavefunction and $\mathcal{H}$ is an operator representing the cncrgy of the system If $\mathcal{H}$ is time-indcpendcnt, then

$$\psi(t) = u\exp\left(\frac{iEt}{\hbar}\right) \tag{9.30}$$

where the wave functions u nrc time-independent, and $\mathcal{H}_0 u = Eu$ where E is the system energy and $\mathcal{H}_0$ rcprcscnts the time-independent Hamiltonian.

The Hamiltonian of a nucleus of $I = \frac{1}{2}$ placed in a magnetic field H is

$$\mathcal{H}_0 = -\gamma\hbar\vec{I}\cdot\vec{H} \tag{9.31}$$

and if $H_0 \| z$, then $\mathcal{H}_0 = -\gamma\hbar H_z I_z$. The energy, E_m, in a magnetic field is given by

$$E_m = -\gamma\hbar H_z m \tag{9.32}$$

where $m = \pm\frac{1}{2}$. The two states correspond to spins parallel and antiparallel to H_0. The Bohr condition gives the natural frequency of the system

$$\frac{E_{-1/2} - E_{+1/2}}{\hbar} = \gamma H = \omega_0 \tag{9.33}$$

Application of an oscillating field requires the solution to the time dependent Schrodinger equation which can be treated as the sum of a time-independent Hamiltonian $\mathcal{H}_0$ and a time-dependent Hamiltonian $\mathcal{H}_1$. The time-independent solution is $\mathcal{H}_0 u_n = E_n u_n$. The wave function $\psi(t)$ can be expanded using time-dependent coefficients a.(t) in conjunction with the stationary wave functions, u_n,

$$\psi(t) = \sum_n a_n(t) u_n \exp\left(-\frac{iE_n t}{\hbar}\right) \tag{9.34}$$

Then the time-dependent Schrodinger equation becomes

$$\begin{aligned} i\hbar\sum_n \frac{\partial a_n}{\partial t} u_n \exp\left(-\frac{iE_n t}{\hbar}\right) \\ = \sum_n a_n \mathcal{H}_1 u_n \exp\left(-\frac{iE_n t}{\hbar}\right) \end{aligned} \tag{9.35}$$

Multiplying equation (9.35) by u_m and integrating gives

$$\frac{\partial a_m}{\partial t} = \frac{1}{i\hbar}\sum_n a_n \mathcal{H}_{mn}\exp(i\omega_{mn} t) \tag{9.36}$$

where $\omega_{mn} = \dfrac{E_m - E_n}{t}$

and $\mathcal{H}_{mn} = (u_m^* | \mathcal{H}_1 | u_n)$

Thw time-dependent part of the Hamiltonian is given by

$$\begin{aligned} \mathcal{H}_1 &= -\hbar\gamma H_1(I_x\cos\omega t - I_y\sin\omega t) \\ &= -\tfrac{1}{2}\hbar\gamma H_1[I_+\exp i\omega t + I_-\exp(-i\omega t)] \end{aligned} \tag{9.37}$$

where $I_\pm$, the raising and lowering operators, are $I_x \pm iI_y$. This equation corresponds to the oscillating field of magnitude H_1 rotating with frequency

ω in the xy plane.

Then $$\frac{\partial a_{+\frac{1}{2}}}{dt} = \tfrac{1}{2} i\gamma H_1 a_{-\frac{1}{2}} \exp[i(\omega_0 - \omega)t]$$

and $$\frac{\partial a_{-\frac{1}{2}}}{\partial t} = \tfrac{1}{2} i\gamma H_1 a_{+\frac{1}{2}} \exp[-i(\omega_0 - \omega)t] \quad (9.38)$$

Taking the second derivative, the differential equation involving $a_{-\frac{1}{2}}$ is obtained,

$$\frac{\partial^2 a_{-\frac{1}{2}}}{\partial t^2} + i(\omega_0 - \omega)\frac{\partial a_{-\frac{1}{2}}}{\partial t} + \frac{1}{4}\gamma^2 H_1^2 a_{-\frac{1}{2}} = 0$$

The solution of this cquntion is

$$a_{-\frac{1}{2}} = A \exp(iP_1 t) + B \exp(iP_2 t) \quad (9.39)$$

where P_1 and P_2 arc dctcrmincd from

$$-P^2 - (\omega_0 - \omega)P + \tfrac{1}{4}\gamma^2 H_1^2 = 0$$

yielding

$$P_1 = -\tfrac{1}{2}(\omega_0 - \omega) + \tfrac{1}{2}[(\omega_0 - \omega)^2 + \gamma^2 H_1^2]^{\frac{1}{2}}$$

and

$$P_2 = -\tfrac{1}{2}(\omega_0 - \omega) - \tfrac{1}{2}[(\omega_0 - \omega)^2 + \gamma^2 H_1^2]^{\frac{1}{2}}$$

For the conditions $t = 0$ and $a_{-\frac{1}{2}} = 0$, B = -A; hence

$$a_{-\frac{1}{2}} = A \exp\left[-\frac{i(\omega_0 - \omega)t}{2}\right] \sin\left(\gamma H_{eff}\frac{t}{2}\right) \quad (9.40)$$

where

$$\gamma H_{eff} = [(\omega_0 - \omega)^2 + \gamma^2 H_1^2]^{\frac{1}{2}}$$

The normsliration condition

$$|a_{+\frac{1}{2}}|^2 + |a_{-\frac{1}{2}}|^2 = 1$$

determines A. Since;

$$|a_{+\frac{1}{2}}|^2 = a_{+\frac{1}{2}}^* a_{+\frac{1}{2}} = \frac{|A|^2}{\gamma^2 H_1^2}\left[(\omega_0 - \omega)^2 + \gamma^2 H_1^2 - \gamma^2 H_1^2 \sin^2\left(\gamma H_{eff}\frac{t}{2}\right)\right]^{\frac{1}{2}}$$

then $$A = \frac{\gamma H_1^2}{[(\omega_0 - \omega)^2 + \gamma^2 H_1^2]^{\frac{1}{2}}}$$

and

$$a_{-\frac{1}{2}} = \frac{\gamma H_1^2 \exp\left[-i(\omega_0 - \omega)\frac{t}{2}\right] \sin\left(\gamma H_{eff}\frac{t}{2}\right)}{[(\omega_0 - \omega)^2 + \gamma^2 H_1^2]^{\frac{1}{2}}} \quad (9.41)$$

Now the total probability, $P_{-\frac{1}{2}}$, of finding the nucleus in the $-\frac{1}{2}$ level is $|a_{-\frac{1}{2}}|^2 = a_{-\frac{1}{2}}^* a_{-\frac{1}{2}}$,

$$P_{-\frac{1}{2}} = \frac{\gamma^2 H_1^2 \sin^2\left(\gamma H_{eff}\frac{t}{2}\right)}{(\omega_0 - \omega)^2 + \gamma^2 H_1^2} \quad (9.42)$$

This equation is identical with the classical solution [Eq. (9.27)]. This development can be extended to higher-order spin systems.

The transition probability between two energy states m and n is given by $P_{mn} \propto |(\psi_m|I_x|\psi_n)|^2$ which is nonzero only when $m = n \pm 1$; thus the first-order selection rule for nuclear levels is $\Delta m = \pm 1$ [38, 79, 81, 82]. The first-order selection rule for many electronic Zeeman transitions is also $\Delta m = \pm 1$; however, other effects, e.g., cyrstal electric field, sufficiently mix the wave functions of the levels, thereby permitting $\Delta m > \pm 1$ transitions to occur.

5. Spin Lattices and Spin-Spin Relaxation

Both the classical and quantum mechanical solutions for. magnetic resonance predict sharp absorption and emission lines. Actual lines are broadened by either homogeneous or inhomogeneous mechanisms. Homogeneous broadening mechanisms include:

1. Dipole-dipole interactions between like spins
2. Spin-lattice relaxation
3. Notional narrowing by local field fluctuations
4. Spin interactions with the exciting field
5. Diffusion of excitation through the spin system.

Examples of inhomogcncous broadening mechanisms are:

1. ESR hyperfine interactions
2. Anisotropic broadening
3. Dipolar and electric quadropolar coupling to unlike spins

The homogeneous broadening mechanisms (1) and (2) act as magnetization damping terms, and their concepts in **terms** of a nuclear spin system are presented below.

Spin-lattice Relaxation. Nuclear spins interact. **with** their surroundings, but the magnitude of the interaction usually permits distinguishing between the temperature of the spin system and the lattice (bath) temperature. Absorption of energy at resonance tends to equalize the populations of the two levels involved, but heat flow from the spin system to the lattice via lattice vibrations (phonons) restores therma1 equilibrium following resonance.

The Boltzmann distribution of a spin system in a magnetic field, H_0, is characterized by a spin temperature T_s [83, 84] such that

$$\frac{N_n}{N_m} = \exp\left(-\frac{E_m - E_n}{kT_s}\right) \quad (9.43)$$

where N_n and N_m are the populations of the nth and mth nuclear Zeeman levels. The magnetization, M_0, at thermal equilibrium as given by Curie's law.

$$M_0 = \frac{CH_0}{T_l} \quad (3.44)$$

where $C = N[\gamma^2 h^2 I(I + 1)/3k]$ defines the lattice tempcraturc, T_l. When $T_s \neq T_l$ a semiequilibrium population distribution occurs where the spin system has internal equilibrium, but it is not in thermal equilibrium with the lattice; hanco $M = CH_0/T_s$. A positive spin temperature occurs when M_s is parallel to H_0 and the level populations decrease with increasing energy. When M_s is antiparallel to H_0, the spin temperature is negative, indicating that the populations of the Zeeman levels increase with increasing energy. This situation is possible since antiparallel

magnetization gives the state of greatest internal energy to the spin system. In many instances, the nuclear magnetization growth or decay rate is a single exponential and the time constant, T_1, is termed the spin-lattice relaxation time. Thus

$$\left(\frac{dM_z}{dt}\right) = \left[\frac{(M_0 - M_z)}{T_1}\right]$$

or in terms of the spin temperature

$$\frac{d}{dt}\left(\frac{1}{T_s}\right) = \frac{1}{T_1}\left(\frac{1}{T_l} - \frac{1}{T_s}\right). \qquad (9.45)$$

The spin-lattice relaxation process can be described by an increase or decrease of a spin system temperature as thermal equilibrium is approached.

Another way of developing the spin-lattice relaxation involves the rate of change of the population difference between two Zeeman levels. Consider a nuclear spin of ½ in a magnetic field H_0. Let W_- rcprcscnt the probability of $-\rightarrow+$ transition, W_+ is the probability of $+\rightarrow-$ transition, and N_-, N_+ are the spin populations of- the $m_I = -\frac{1}{2}$ and $m_I = \frac{1}{2}$ respectively; then according to the principle of detail balancing $N_+W_+ = N_-W_-$ The level populations N_+, N_- are given by the Boitzmann distribution

$$\frac{N_-}{N_+} = \exp - \frac{\Delta E}{kT} \qquad (9.46)$$

Therefore,
$$\frac{W_+}{W_-} = \exp - \frac{\Delta E}{kT} \qquad (9.47)$$

or the probability of emission exceeds the absorption probability. The population difference n, is simply $(N_+ - N_-)$ and its rate of change is given by

$$\frac{dn}{dt} = 2N_-W_- - 2N_+W_+$$

$$= 2W_+\left[N_-\left(I + \frac{\gamma\hbar H_0}{kT}\right) - N_+\right] \qquad (9.48)$$

Then to first order

$$\frac{\gamma\hbar H_0}{kT}N_- = \frac{N_0}{2}\frac{\gamma\hbar H_0}{kT} = n_0 \qquad (9.49)$$

where n_0 is the population difference when thermal equilibrium between the lattice and spin system exists. Now

$$\frac{dn}{dt} = 2W_+(n_0 - n) \qquad (9.50)$$

which has the solution

$$n_0 - n = (n_0 - n_i)\exp(-2W_+t) \qquad (9.51)$$

where n_i is the initial value for n. The time constant $(2W)^{-1}$ is defined as the spin-lattice relaxation time, T_1. Thcrmnl or longitudinal relaxation have been used interchangeably with spin-lattice relaxation.

The spin-lattice relaxation time can be writtcn in terms of the magnetization by multiplying (9.51) by μ,

$$\mu\frac{dn}{dt} = 2W_+\mu\left[N_-\left(1 + \frac{\mu H_0}{kT}\right) - N_+\right] \qquad (9.52)$$

Since $\mu(N_+ - N_-) = \mu n = M_z$

and $(N_+ + N_-)\frac{\mu^2 H_0}{kT} = M_0$

then the fundamental equation results,

$$\frac{dM_z}{dt} = 2W(M_0 - M_z) \qquad (9.53)$$

where $T_1 = (2W)^{-1}$ and W is the transition probability for absorption.

Spin-Spin Relaxation. Each magnetic moment interacts with the local magnetic field,

$$\left[H_l \simeq \frac{\mu}{r^3}(3\cos^2\theta - 1)\right] \qquad (9.54)$$

produccd by neighboring nuclei. For hydrogen ($\mu = 1.41 \times 10^{-23}$ cm³ G) this local field is 14 gauss for a nearest neighbor distance of 1 A. In the presencc of an external magnetic field, H_z, the magnetic moment of nucleus f in a solid directed parallel to H_z will be constant. if the time-dependent effects produccd by precessing neighbor moments g are ignored. The mean-square local field at the f nucleus due to the remaining nuclei is

$$\left\langle \sum_g (H_{zg}^2) \right\rangle = \sum_g \frac{\langle \mu_{zg}^2 \rangle (1 - 3\cos^2\theta_{fg})^2}{r_{fg}^6} \qquad (9.55)$$

If all g nuclei are alike but differ from nucleus f,

$$\langle \mu_{fg}^2 \rangle = \tfrac{1}{3}I(I+1)\gamma^2\hbar^2$$

and $\langle (\Delta H)^2 \rangle = \langle (H - H_z)^2 \rangle$

$$= \tfrac{1}{3}I(I+1)\gamma^2\hbar^2 \sum_g \frac{(1 - 3\cos^2\theta_{fg})^2}{r_{fg}^6} \qquad (9.56)$$

when nuclei g and nucleus f have the same moment. Then Eq. (9.56) must. be multiplied by ¾ since a nucleus g precessing in the field H_z, gives a resonant time-dependent field at nucleus f and vice versa, with the result that simultaneous spin flips, $\Delta m_f = \pm 1$ and $\Delta m_g = \mp 1$, occur. This moment reorientation (flip-flop) conserves total energy and limits the lifetime of the spin in an energy Icvel, thcrehy contributing to the line broadening.

The precession frequency for $\langle H_l \rangle$ from the Larmor equation is

$$\Delta\omega = \frac{\mu\langle H_l \rangle}{I\hbar}$$

If the nuclear moments are precessing in phase at a time t, then the time $\Delta\omega^{-1}$ is required for the moments to get out of phase. These local field effects limit the phase memory lifetime of a spin state. Hence, the spin-spin or transverse relaxation time is defined as the characteristic time T_2, required for the precessing spins to lose phase. Since local field variations give a range of absorption frequencies a line width of the order of H_{local} is obtained. Usually the line shape function $g(\omega)$ is defined Such that $T_2 = \pi g(\omega)_{max}$.

6. Bloch Phenomenological Equations

Bloch [86, 87] introduced a set of phenomenological equations which describe the interaction of the macro-

scopic magnetization with a **linear** oscillating field. These classical equations are rigorous for many experimental conditions e.g., transient effects [88], spin echoes [89], and other time-dependent phenomena in both nuclear and electron magnetic resonance

The rate of change of the components of nuclear magnetization in a static magnetic field, H_0, applied in the z direction and a linear oscillating field, H_1, applied perpendicular to H_0 are

$$\begin{aligned}\frac{dM_x}{dt} &= \gamma(M_yH_z - M_zH_y)\\ \frac{dM_y}{dt} &= \gamma(M_zH_x - M_xH_z)\\ \frac{dM_z}{dt} &= \gamma(M_xH_y - M_yH_x)\end{aligned} \tag{9.57}$$

The components of the magnetic field are

$$\begin{aligned}H_z &= H_0\\ H_x &= H_1\cos\omega t\\ H_y &= H_1\sin\omega t\end{aligned}$$

Hence

$$\begin{aligned}\frac{dM_z}{dt} &= \gamma(M_yH_0 + M_zH_1\sin\omega t)\\ \frac{dM_y}{dt} &= \gamma(M_zH_1\cos\omega t - M_xH_0)\\ \frac{dM_z}{dt} &= -\gamma(M_xH_1\sin\omega t + M_yH_1\cos\omega t)\end{aligned} \tag{9.58}$$

Bloch introduced two damping terms, the spin-lattice relaxation time T_1 and the spin-spin relaxation time T_2. The spin-spin relaxation causes the M_x and M_y components to decay whereas spin-lattice relaxation affects the M_z component. Thus the complete Bloch equations are

$$\begin{aligned}\frac{dM_x}{dt} &= \gamma(M_yH_0 + M_zH_1\sin\omega t) - \frac{M_x}{T_2}\\ \frac{dM_y}{dt} &= \gamma(M_zH_1\cos\omega t - M_xH_0) - \frac{M_y}{T_2}\\ \frac{dM_z}{dt} &= -\gamma(M_xH_1\sin\omega t + M_yH_1\cos\omega t) - \frac{M_z - M_0}{T_1}\end{aligned} \tag{9.59}$$

These equations take a simpler form when the coordinate reference system rotating with the applied field H_1 is employed. In this system u and v are defined as the components of M parallel (in phase) and perpendicular (out of phase) to the direction of H_1. Using the relations between the components

$$\begin{aligned}u &= M_x\cos\omega t - M_y\sin\omega t\\ v &= -M_x\sin\omega t - M_y\cos\omega t\end{aligned} \tag{9.60}$$

the Bloch equations in the rotating frame become

$$\begin{aligned}\frac{du}{dt} + \frac{u}{T_2} + (\omega_0 - \omega)v &= 0\\ \frac{dv}{dt} + \frac{v}{T_2} - (\omega_0 - \omega)u + \gamma H_1M_z &= 0\\ \frac{dM_z}{dt} + \frac{M_z - M_0}{T_1} - \gamma H_1v &= 0\end{aligned} \tag{9.61}$$

The steady-state solutions for these equations are

$$\begin{aligned}u &= M_0\frac{\gamma H_1T_2^2(\omega_0 - \omega)}{1 + T_2^2(\omega_0 - \omega)^2 + \gamma^2H_1^2T_1T_2}\\ v &= -M_0\frac{\gamma H_1T_2}{1 + T_2^2(\omega_0 - \omega)^2 + \gamma^2H_1^2T_1T_2}\\ M_z &= M_0\frac{1 + T_2^2(\omega_0 - \omega)^2}{1 + T_2^2(\omega_0 - \omega)^2 + \gamma^2H_1^2T_1T_2}\\ M_x &= \tfrac{1}{2}M_0\gamma T_2^2\frac{(\omega_0 - \omega)2H_1\cos\omega t + 2H_1\sin\omega t}{1 + T_2^2(\omega_0 - \omega)^2 + \gamma^2H_1^2T_1T_2}\\ M_y &= \tfrac{1}{2}M_0\gamma T_2^2\frac{2H_1\cos\omega t - T_2(\omega_0 - \omega)2H_1\sin\omega t}{1 + T_2^2(\omega_0 - \omega)^2 + \gamma^2H_1^2T_1T_2}\end{aligned} \tag{9.62}$$

Since both M_x and M_y contain a term in phase with $H_1(2H_1\cos\omega t)$ and a term out of phase with $H_1(2H_1\sin\omega t)$, Bloch suggested the use of complex susceptibility, $\chi = \chi' - i\chi''$.

Thus

$$\begin{aligned}M_x &= \chi'(2H_1\cos\omega t) + \chi''(2H_1\sin\omega t)\\ M_y &= \chi'(2H_1\cos\omega t) - \chi''(2H_1\sin\omega t)\end{aligned} \tag{9.63}$$

where

$$\begin{aligned}\chi' &= \tfrac{1}{2}\chi_0T_2\omega_0\frac{T_2(\omega_0 - \omega)}{1 + T_2^2(\omega_0 - \omega)^2 + \gamma^2H_1^2T_1T_2}\\ \chi'' &= \tfrac{1}{2}\chi_0T_2\omega_0\frac{1}{1 + T_2^2(\omega_0 - \omega)^2 + \gamma^2H_1^2T_1T_2}\end{aligned}$$

Figure 9.5 shows the dispersion χ' and absorption χ'' components as functions of the dimensionless product

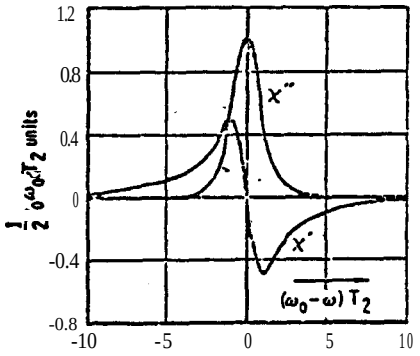


FIG. 9.5. Absorption (χ'') and dispersion (χ') Lorentzian line shapes.

$T_2(\omega_0 - \omega)$. It the spin temperature is equal to the lattice temperature, then χ'' can bc written in terms of the line function $g(\omega)$ as

$$\chi'' = \tfrac{1}{4}\chi_0\omega_0g(\omega) \tag{9.64}$$

Thus

$$g(\omega) = \frac{2T_2}{1 + T_2^2(\omega_0 - \omega)^2}$$

which is the Lorentzian line shape characteristic of a damped harmonic oscillator. Line shapes arc **not** always approximated by a Lorentzian-type curve and hence limits the Bloch formalism. The spin-lattice relaxation theory (90) predicts a Lorentzian line shape but the spin-spin relaxation theory (91) indicates a Gaussian line shape.

Homogeneous broadening yields **Lorentzian** line shapes whereas Gaussian line shapes occur when inhomogenous broadening is involved. Some mag-

netic resonance line shapcs can be fitted by a combination of the Lorentzian and Gaussian functions.

The rate of energy absorption per unit volume, A, is given by

$$A = \frac{\omega}{2\pi}\int_0^{2\pi} 2H_1 \cos\omega t \left(\frac{dM_x}{dt}\right) dt$$

$$= \frac{\omega}{2\pi}\int_0^{2\pi} 4H_1^2\omega \cos\omega t\{-\chi' \sin\omega t + \chi'' \cos\omega t\}\, dt$$

$$= 2\omega H_1^2\chi'' \qquad (9.65)$$

This equation indicates that energy absorbed by the spin system from the oscillating field is proportional to the χ'' magnetization component.

The conditions under which the Bloch equations apply [87] are:

1. The effect of all other spins upon a given spin is equivalent to that of a "bath" in thermodynamic cquiiibrium.
2. The "bath" temperature remains constant.
3. $\gamma H_1, T_1^{-1}, T_2^{-1} \ll \gamma H_0$ and γH_{local}.
4. The spin-bath interaction can be expanded in multipoles.
5. When two unpaircd spins occur, $kT \gg \hbar\omega_0$, or for more than two unpaired spins, the quadrupole coupling, $eqQ, \ll \hbar\omega_0$.

The steady-state solution of the Bloch equations assumes that thermal equilibrium exists between the magnetization and the oscillating field. Thus the rate of passage through resonance influences the resulting line shape. The slow-passage condition $(d/dt)[(H_0 - H)/H_1] \ll (T_1T_2)^{-\frac{1}{2}}$ is required for the steady-state solution and the true line shape is observed when $dH/dt \ll \gamma(\Delta H)^2$, where ΔH is the line width at half maximum. The adiabatic rapid-passage condition, where $\gamma H_1 \gg (d/dt)[(H_0 - H)/H_1] \gg (T_1T_2)^{-\frac{1}{2}}$, may be employed to observe weak signals or long relaxation times. Other transient response effects which occur during resonance passage have been described in detail [89, 92-96].

When $\gamma^2H_1^2T_1T_2 \approx 1$, the resonance width increases and the resonance height **decreases;** at sufficiently high amplitudes of H_1 no resonance absorption is observed because the energy-level populations have been equalized or the resonance is saturated. The term $1 + \gamma^2H_1^2T_1T_2$ is called the saturation factor. The absorption mode χ'' saturates rapidly when $H_1 \geq (\gamma T_1T_2)^{-\frac{1}{2}}$, but the dispersion mode requires higher H_1 values for saturation [97].

7. Magnetic Resonance Spectrometers

The essential components of a magnetic resonance spectrometer are (1) 3 magnet to provide H_0, (2) a generator at the appropriate frequency to provide H_1, and (3) a detector to indicate the resonant condition.

Permanent magnets [98, 99], electromagnetis [43], and superconducting solenoids [100] have been used in magnetic resonance spcctromctcrs and the particular choice depends on the intended application. High-resolution NMR spectral measurements requiring extremely high homogeneity can be made with any of the above magnets; however, thc electromagnet is preferred for ESR measurement where large field variations are desired. Significant signal-strength enhancement can bc obtained with higher magnetic fields; therefore, the high field strengths which can be attained with superconducting solenoids may be used for low abundant nuclides or low spin concentrations. To traverse the resonance line, either frequency sweeping [101] or magnetic-field sweeping [43] as a function of time is required.

When signal-to-noise improvement is required e.g., wide lines or low spin concentrations, a small sinusoidal modulation, ω_m, is applied to the magnetic field [102]; then

$$H = Ho + H_m \cos\omega t$$

For wide NMR lines $\omega_m \approx 10\text{–}10^2$ hertz whereas $10^2\text{–}10^5$ hertz is employed in EPR spectroscopy. In conjunction with magnetic-field modulation, a narrow-band amplifier and phase-sensitive detector are used. In this case, the first derivative, $d\chi'/dH_0$ or $d\chi''/dH_0$, is obtained.

NMR spectrometers can be divided into three major types: the double-coil type, the bridge type, and the marginal oscillator. The double-coil NMR spectrometer [30] shown in Fig. 9.6 is widely used.

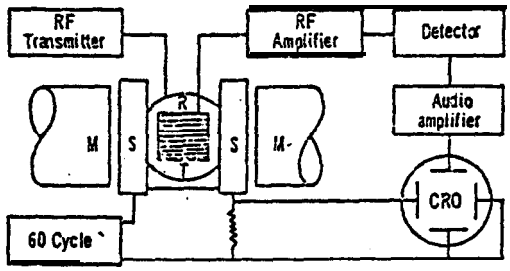


FIG. 9.6. Schematic diagram of the double-coil spectrometer used by Bloch, Hansen, and Packard [30]. M = magnet, S = sweep coils, T = transmitter coil, R = receiver coil.

In this spectrometer, the sample is placed inside the receiver coil with its axis perpendicular to both the transmitter coil and to the magnetic-field direction. Control of the radio-frequency leakage from the transmitter coil permits selection of either the absorption or dispersion magnetization component. Off resonance, no voltage is induced in the receiver coil, but at resonance the radiation emitted by the excited nuclei returning to lower energy levels induces a voltage E_r in the n turn reciver coil,

$$E_r = 4\pi An\frac{dM_y}{dt}$$

where A is the sample cross section. This voltage change is diode-detected and amplified for oscilloscope or recorder presentation.

In the bridge-type [29] NMR spectrometer a single inductance coil in a tuned LC circuit serves as both the transmitter and receiver coil. Off resonance, the radio-frequency voltage arriving at the amplifier is balanced out with an r-f bridge. Susceptibility component selection is made by unbalancing the bridge in the proper phase. This bridge unbalance must be significantly larger than the resonance signal for pure

mode selection. At resonance the energy absorbed from the tuned circuit by the sample further unbalances the bridge and gives a voltage identical to that of the double-coil spectrometer. A double-bridge NMR spectrometer [103] providing both amplitude and phase balance minimizes fluctuations associated with large H_1 levels and gives better mode selection.

The sample coil of the marginal oscillator spectrometer [101-106] forms part of a tuned circuit, which controls the oscillator frequency by regnerative feedback. In the marginal condition the feedback circuit time constant is sufficient to maintain nearly constant oscillation amplitude; however, at resonance, the long time constant does not permit the generator to follow the energy ahsorption. The great advantages of this spectrometer are pure absorption mode detection, high sensitivity, and the case of frequency changes.

ESR spectrometers have operated in the frequency range 1 MHz [107, 108] to 158.5 GHz [109]. However, the usual commercial spectrometers operate at X band (8.2-12.4 GHz), K band (18-26.5 GHz), and K_a band (26.5-40 GHz) or Q band (33-50 GHz) because of increased sensitivity and higher resolution at microwave frcqucncics.

Two major types [12, 15, 16, 110] of ESR spectrometers, transmission and reflection, are used. In the transmission spectrometer, the sample is placed in the waveguide between the microwave generator (usually a reflex klystron) and the detector. At resonance, the microwave energy absorption is determined with a simple silicon crystal diode or bolometcr.

A bock diagram of a reflection-type ESR spectrometer is shown in Fig. 9.7. Microwaves pass from the stabilized klystron, through arm 1 of the magic tee to a high quality factor reflection cavity in which the sample is located. By mechanically and electronically tuning the frequency of the klystron, its operating mode can bc matched to the resonant frequency of the reflection cavity. After being properly erly tuned to the cavity, the klystron is frequency- or phase-stabilized at the cavity resonant frequency. Off resonance the cavity will have a representative quality factor, Q, which is given by

$$Q = \frac{\omega[\text{total stored energy}]}{\text{power lost per cycle}} = \frac{\omega L}{R}$$

At resonnncc additional power is lost to the sample resulting in a decreased Q. This decrease, dQ, is given by

$$dQ = -4\pi\eta Q^2\chi''$$

where η is the filling factor which is the fraction of the coil volume occupied by the snmplc. Three general types of detectors [111-113] can be used (1) bolometer, (2) crystal diode, and (3) superheterodyne, in the third arm of the magic tee to detect the resonance. Usually additional amplification is provided before the resonance is recorded.

The experimental methods for measuring relaxation processes are basically the same for both NMR [40, 44, 50, 56, 114, 115] and EPR [116-120]. These methods can be divided into saturation methods and pulse methods. The saturation methods can be divided into three techniques: the direct, the progressive or steady-state saturation, and the polarization reversal. In the direct method [44] H_1 is increased to produce saturation and then suddenly it is reduced to a small value. The observed signal recovery follows a single exponential $[1 - \exp(-t/T_1)]$ in many cases.

The progressive-saturation technique [44] depends on the saturation factor $(1 + \gamma^2 H_1^2 T_1 T_2)^{-1}$. At a low H_1 value T_2 is evaluated from the line width. The signal amplitude as a function of H_1 decreases

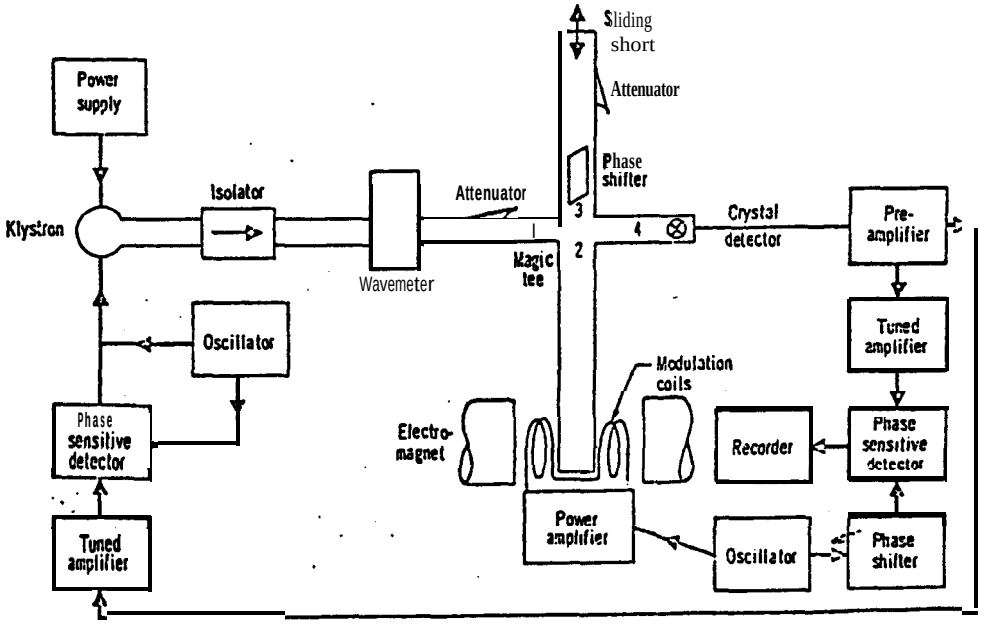


FIG. 9.7. Block diagram of a typical reflection cavity electron spin resonance spectrometer.

rapidly when $\gamma^2 H_1^2 T_1 T_2 \gg 1$. Then T_1 can be evaluated from T_2 and the value of H_1 at the sample.

The polarization-reversal method [114-121] employs an adiabatic fast passage through resonance plus a large H_1 field to reverse M_z from the z to $-z$ direction. At equilibrium the return signal $(-d\chi''/dH)$ will have the opposite sign and the same amplitude as the forward signal $(d\chi''/dH)$. The signal amplitude depends on the time above and below resonance and is proportional to the magnetization at each passage. Thus T_1 can be evaluated from the signal amplitudes at two different sweep frequencies or by using unequal time intervals above and below resonance.

Two pulse methods, often referred to as the incoherent and coherent techniques, are used for accurate relaxation-time measurements. Incoherent techniques usually refer to relaxation measurements which invoke magnetization recovery from single high-power H_1 pulses. Most electron spin relaxation measurements are performed with the saturation recovers method [116, 117] because of experimental difficulties associated with a microwave spin echo spectrometer [119, 120].

The spin echo method 89] is an example of a coherent technique for relaxation measurements. In this method the magnetization is rotated away from the direction of static field with a timed series of pulses of amplitude and duration such that

$$\gamma H_1 t_w = \frac{\pi}{\theta} \quad (90^\circ \text{ pulse})$$

[89] or a combination of a 90° pulse and a 180° pulse [122]. In the latter case a signal (the echo) will occur at time 2τ where τ is the time interval between pulses. The echo amplitude, A, is given by

$$A = \exp\left(-\frac{2\tau}{T_2} - k\frac{(2\tau)^3}{3}\right)$$

where $k = \frac{1}{4}\gamma^2 G^2 D$, with D being the diffusion coefficient and G being the magnetic-field gradient. T_2 can be obtained from a $180^\circ - 90^\circ$ sequence [122] or a series of 90° pulses spaced at variable time intervals [89, 123].

8. Characteristics of NMR Spectra

Solids. Nuclear resonances of solids are both homogeneously and inhomogeneously broadened. The major line broadening mechanisms are:

1. Dipole-dipole
2. Electric qundrupole
3. Electron-coupled spin-spin interactions

For $I = \frac{1}{2}$, the dipole-dipole coupling gives an H_z component at each nucleus of $H_0 \pm H_l$, where H_l is the local magnetic field [Eq. (9.66)] caused by neighboring nuclei. This magnetic [36, 124] intersection has the form

$$\mathcal{H}_d = \sum_{j>k} \frac{[\vec{\mu}_j \cdot \vec{\mu}_k - 3(\vec{\mu}_j \cdot \vec{r}_{jk})(\vec{\mu}_k \cdot \vec{r}_{jk})]}{r^3_{jk}} \quad (9.66)$$

and predicts that for a single crystal sample having isolated pairs of nuclei two equally intense resonance lines separated by $3\mu/r^3(3\cos^2\theta - 1)$ occur. This approach also applies to polycrystalline samples although averaging over all θ values causes appreciable resonance overlap. The prediction of line shapes for groups of three nuclei at the corncrs of an isosceles triangle [125] and a tetrahedral arrangement of nuclei [36, 47, 126, 127] has been experimentally verified.

When more complicated grouping of nuclei occur, the NMR line often is unresolved. Useful information can be obtained from this structureless line by the Van Vleck moment method [85] which expresses the internuclear distances in crystals in terms of the second and fourth moment. The general second mqmcnt for both like and unlike spins is

$$\langle(\Delta H)^2\rangle_{\text{ave}} = \frac{3}{4}\gamma_i^4 \frac{I_i(I_i+1)}{N}\hbar^2 \sum_{i>j}(3\cos^2\theta_{ij} - 1)^2 r_{ij}^{-6} + \frac{1}{3}\frac{\gamma_i^2\gamma_j^2}{N} I_{j'}(I_{j'}+1)\hbar^2 \sum_{i,j'}(3\cos^2\theta_{ij'} - 1)^2 r_{ij}^{-6} \quad (9.67)$$

Bond distances in crystals can be evaluated when one bond length of the crystal is unknown [36, 47, 127]. When an axial symmetric electric field gradient $\partial^2 V/\partial z^2$ due to nuclear electric quadrupolsr coupling eQ, exists at the nucleus, the interaction energy is

$$\frac{eQ}{4I(2I-1)}\frac{\partial^2 V}{\partial z^2}[3m^2 - I(I+1)] \quad (9.68)$$

which must be added to the magnetic field nuclear moment energy [69, 73]. This electric field gradient introduces asymmetry in the nuclear Zeeman splittings, thereby allowing observation of $2I$ resonances instead of the single resonance characteristic of nuclei on which the quadrupole moment is relaxed.

The effective second moment in solids is reduced by motion of lattice points about an equilibrium position as well as molecular group vibration and rotation [20, 26, 46]. For free rotation a time average of all θ values (Eq. (9.67)] over all azimuthal angles is made. This averaging process increases the fourth moment but does not affect the second moment [128]. When the line width is much smaller than the rotation frequency the resonance tails become broadened beyond detection and an effective second-moment decrease results. Motional narrowing induced by increased temperature has been used to study phase transformations, tunneling, and hindered rotational barriers. In addition rapid rotation of solid samples at an axis inclined at $\cos^{-1} 1/\sqrt{3}$ to H_0 provides motional narrowing which permits resolution of nonequivalent site resonances [120-131]. The theory of motional narrowing in solids is directly applicable to the random Brownian motion of liquids [103].

Although rapid random motion averages the direct dipole-dipole coupling to zero, the resonance frequencies for identical nuclei located in chemically

nonequivalent sites differ. Differences in the time average magnetic field at the nucleus are caused by

1. Interactions of the nucleus with the electrons of the same atom or molecular species
2. Nuclear interactions with other atoms or molecular species
3. Bulk susceptibility effects

Thus these shielding mechanisms depend on the chemical environment of the nucleus and hence resonance position differences between nonequivalent sites are called chemical shifts [132, 133]. The interaction Hamiltonian for a set of nuclei γ_i is given by

$$\mathcal{H}_0 = \sum_i \gamma_i H_i I_z(i) \tag{9.69}$$

Calculations of atomic shielding constants agree favorably with experimental resonance shifts [50, 134], but calculations of shielding constants for molecules require knowledge of the energies and wave functions of all excited electronic states which arc not usually available [50]. Hence a relative chemical shift, δ, for molecules is used, where

$$\delta = \frac{H_S - H_R}{H_R} \times 10^6 \tag{9.70}$$

H_S is the magnetic field for the sample and H_R is the resonance field for a reference compound. Tables of these relative NMR chemical shifts for many nuclei have been presented [32, 35, 40-42, 50, 59, 63, 133, 136]. Chemical shifts as large as 10^4 parts per million have been measured for diamagnetic cobalt compounds but the usual range is 10-1,000 ppm.

The NMR spectrum of a molecule containing chemically nonequivalent groups, will exhibit multiplet resonance structure caused by indirect spin-spin coupling transmitted between nuclei by the bonding electrons. This indirect coupling may be represented as an additional Hamiltonian involving scalar products Of all pairs of nuclei,

$$\mathcal{H}_1 = \sum_{i<j} J_{ij} \vec{I}_i \cdot \vec{I}_j \tag{9.71}$$

whore J_{ij} is the magnetic-field-independent spin-spin coupling constant. General procedures for the analysis of complex NMR spectra in terms of coupling constants and chemical shifts have been presented [40, 50, 63, 137-140].

Metals. Comparison of the Cu^{63} NMR resonances for Cu and CuCl shows the Cu resonance at a higher magnetic field [141, 142]. This shift is proportional to the field strength but the relative shift

$$\frac{H_{\text{metal}} - H_{\text{reference salt}}}{H_{\text{reference salt}}} = \frac{\Delta H}{H}$$

(the Knight shift) is independent of the field. Furthermore, Knight shifts do not depend on moments of different isotopes. They arise principally from the contact interaction between the nuclear magnetic moment and the spin magnetization field of the s conduction electrons. The shift is proportional to the spin magnetization density [143] or

$$\frac{\Delta H}{H} = \frac{8\pi}{3} \chi_v V_0 \langle |\psi(0)|^2 \rangle_F \tag{9.72}$$

where χ. is the volume spin susceptiblity

$$(\chi. = 2\mu_B^2 N(E_F)$$

$N(E_F)$ is the density of states per unit energy interval at the Fermi surface], V_0 is the atomic volume and $\langle |\psi(0)|^2 \rangle_F$ is the probability density [47] at the nucleus for s conduction electrons of energies E_F averaged over the Fermi surface. Knight-shift measurements for pure solid and liquid metals [45], intermetallics [144], alloys (single-phase, multi-phase, and ordered) [45] and superconductors [145] have provided useful information for characterization [45] of these materials.

Acoustic Wave-Spin Interaction [146-148] One of the mechanisms responsible for spin-lattice relaxation of nuclear systems is the indirect or Raman interaction process which invokes two high-frequency lattice vibrations. In this process one phonon is absorbed by the spin system and another phonon is emitted after de-exciting the spin system, The frequency diffcrcncc between these phonons is $\pm\nu_0$ or $2\nu_0$ where ν_0 is the Larmor frequency. This inelastic phonon scattering process is highly probable since the frcquencics, ν_L, of most of the thermal phonons are larger than the resonant frequency.

Application of a coherent acoustic phonon beam will modulate one of the internal interactions, thereby exchanging energy with the spin system, e.g., coupling 25.33 MHz acoustic waves to $KMnF_3$ causes $\Delta m_I = \pm 1$ transitions in the F^{19} NMR line [149].

9. Characteristics of EPR Spectra

EPR spectra are described by the spin Hamiltonian (4, 150-152] $\mathcal{H}_s$, which is essentially the system energy written in. terms of the system parameters. The spin Hamiltonian can be divided into an electronic part $\mathcal{H}_e$ and a nuclear part $\mathcal{H}_n$. The electronic part $\mathcal{H}_e$ is given by

$$\mathcal{H}_e = \mu_B \vec{H} \cdot g \cdot \vec{S} + \vec{S} \cdot D \cdot \vec{S} \tag{9.73}$$

and the nuclear part $\mathcal{H}_n$ is given by

$$\mathcal{H}_n = \vec{S} \cdot T \cdot \vec{I} + \vec{I} \cdot P \cdot \vec{I} - g_N \mu_N \vec{H} \cdot \vec{I} \tag{9.74}$$

The first term of $\mathcal{H}_e$ describes the Zeeman splitting. The second term of $\mathcal{H}_e$ represents both the magnetic-dipole interaction between electrons and second order effects of spin orbit coupling. The first term of $\mathcal{H}_n$ is the hyperfine interaction and nuclear quadrupolar coupling is represented by the second term. The other terms of $\mathcal{H}_e$ and SC, represent the magnetic-field interactions with the electron and nucleus respectively.

For the case of a single electron with zero orbital angular momentum in a magnetic field,

$$\mathcal{H}_e = \mu_N g H S_z \tag{9.75}$$

with g being near g_0, the spin-only value. Quenching of orbital angular momentum is characteristic of most

organic monoradicals [22-26, 153, 154]. When spin-orbit coupling is prcscnt, mixing of higher orbital states occurs, thereby causing an effective anisotropic magnetic moment. For this case the Zeeman interaction is written in terms of its components,

$$\mathcal{H}_z = \mu_B(g_x S_x H_x + g_y S_y H_y + g_z S_z H_z)$$

where g_x, g_y, and g, are the principal values of the g tensor and S_x, S_y, and S_z refer to effective spin components. In the simple case of axial symmetry in a crystal only two g components, $g_\parallel = g_z$ (parallel component) and $g_\perp = g_x = g_y$ (perpendicular component) are required. The g value in a direction making an angle θ with the symmetry axis is

$$g = (g_\parallel^2 \cos^2\theta + g_\perp^2 \sin^2\theta)$$

Where spin-orbit coupling or crystal field effects are present, g values differing from g_0 are observed [13, 14, 17-19, 21-27].

Initial splittings of the effective spin levels of ions or biradicals in crystals often occur. The strong asymmetric electric field of the crystal interacting through the spin-orbit coupling is largely rcsponsiblc for this zero field splitting. The second psrt of $\mathcal{H}_e$, the fine structure term, $\vec{S} \cdot D \cdot \vec{S}$, expresses this interaction. A twofold degeneracy always exists for a half-integral spin system in a crystal field (Kramer's rule), but for an integral spin system this level degeneracy may be resolved. For an axially symmetric crystal field

$$\vec{S} \cdot D \cdot \vec{S} = D[S_z^2 - \tfrac{1}{3}S(S+1)]$$

and for crystals with lower symmetry

$$\vec{S} \cdot D \cdot \vec{S} = D[S_z^2 - \tfrac{1}{3}S(S+1)] + E(S_x^2 - S_y^2)$$

Thus for $S = \frac{1}{2}$ the zero field splitting between $E_{\pm\frac{1}{2}} - E_{\pm\frac{1}{2}} = 2D$ for an axially symmetric crystal and $2(D^2 + 3E^2)^{\frac{1}{2}}$ for lower symmetry crystals; when S = 1 the axially symmetric zero field splitting, $E_{\pm 1} - E_0 = D$, whereas $E_{+1} - E_{-1} = 2E$ and

$$E_{-1} - E_0 = D - E$$

for positive D and E values when lower crystal symmetrics are involved. In the prcscncc of both an applied magnetic field and a crystal field level crossing occurs and because of wave function mixing $\Delta M > \pm 1$ transitions are allowed.

The hyperfine term of $\mathcal{H}_n$ can be expressed as

$$\vec{S} \cdot T \cdot \vec{I} = AI_zS_z + B(I_xS_x + I_yS_y)$$

where A and B measure the hyperfine splitting parallel and pcrpendiculnr to the crystal axis. The effect of the hyperfine interaction is to split the electron resonance line into $2I + 1$ lines which to first order have the same separation. The resonance multiplicity serves to identify the coupling nucleus or nuclei, and information on the electron density in the molecule [26] is obtained from the separation.

The second term of $\mathcal{H}_n$ represents the nuclear electric quadrupole hyperfine interaction and to first order, it causes no change in the hyperfine resonance positions. However, it often is rcsponsiblc for "forbidden" transitions which appear near the central resonance of the EPR spectrum. When small hyperfine interactions occur, the nuclear Zeeman term may give observable direct field effects by which the nuclear spin flips during an electron resonance [155].

Detailed procedural methods for deriving the spin Hamiltonian parameters from observed EPR spectral data have been developed [4, 6, 14-20, 25-27, 156], and the parameter values for various ions and biradicals have been listed [4-27].

10. Masers

Continuous high-gain microwave amplification by stimulated emission of radiation uses at least three unequally spaced Zeeman levels of a magnetically dilute paramagnetic ion $S \geq 1$ at 4.2°K [157]. In the three-level spin system the populations of the upper and lower energy levels are saturated by high-amplitudc microwaves of the appropriate energy. When a weak microwave signal of energy equal to the separation between the upper and middle energy levels (or the middle and lower energy Ievels) is received by the hot spin system transitions from the higher to the lower energy level occur, thereby amplifying the triggering signal. Microwave signals of other energies do not affect the spin system. Since the separation between Zeeman levels is readily varied with the magnetic field, masers covering a wide frequency range are possible provided proper microwave or other saturating pumps are available. When the Zeeman level separation of the pumped levels approximates the zero field splitting, a decrease in the wave function mixing occurs which decreases the transition probability between the levels. Therefore the frequency tunability of the maser crystal is limited. When the spin-lattice interaction between pump levels is strong (short T_1), these levels will not saturate and thus will limit the amplifier gain.

Two types of solid-state masers, the regenerative reflection cavity maser and the nonregenerative traveling-wave maser have been developed. In the cavity maser, the energy emitted by the maser crystal increases the microwave field in the cavity, thereby increasing the rate of emission from the crystal. Consequently, the maser gain bandwidth product $(G^{\frac{1}{2}}B)$ is constant for the cavity maser. The energy emitted by a slow wave element of a traveling-wave maser increases the microwave field traveling through the structure and thus increases the rate of energy emission from succeeding active elements without reacting with the original element. Therefore, the gain of the traveling-wave maser is a function of its length and the $G^{\frac{1}{2}}B$ is not constant. The noise temperatures for masers are ~10-15°K and electronic gains of ~8 dB/in. are observed for traveling-wave masers, whereas $G^{\frac{1}{2}}B$ products of 40-250 MHz are obtained for cavity masers [158-162].

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